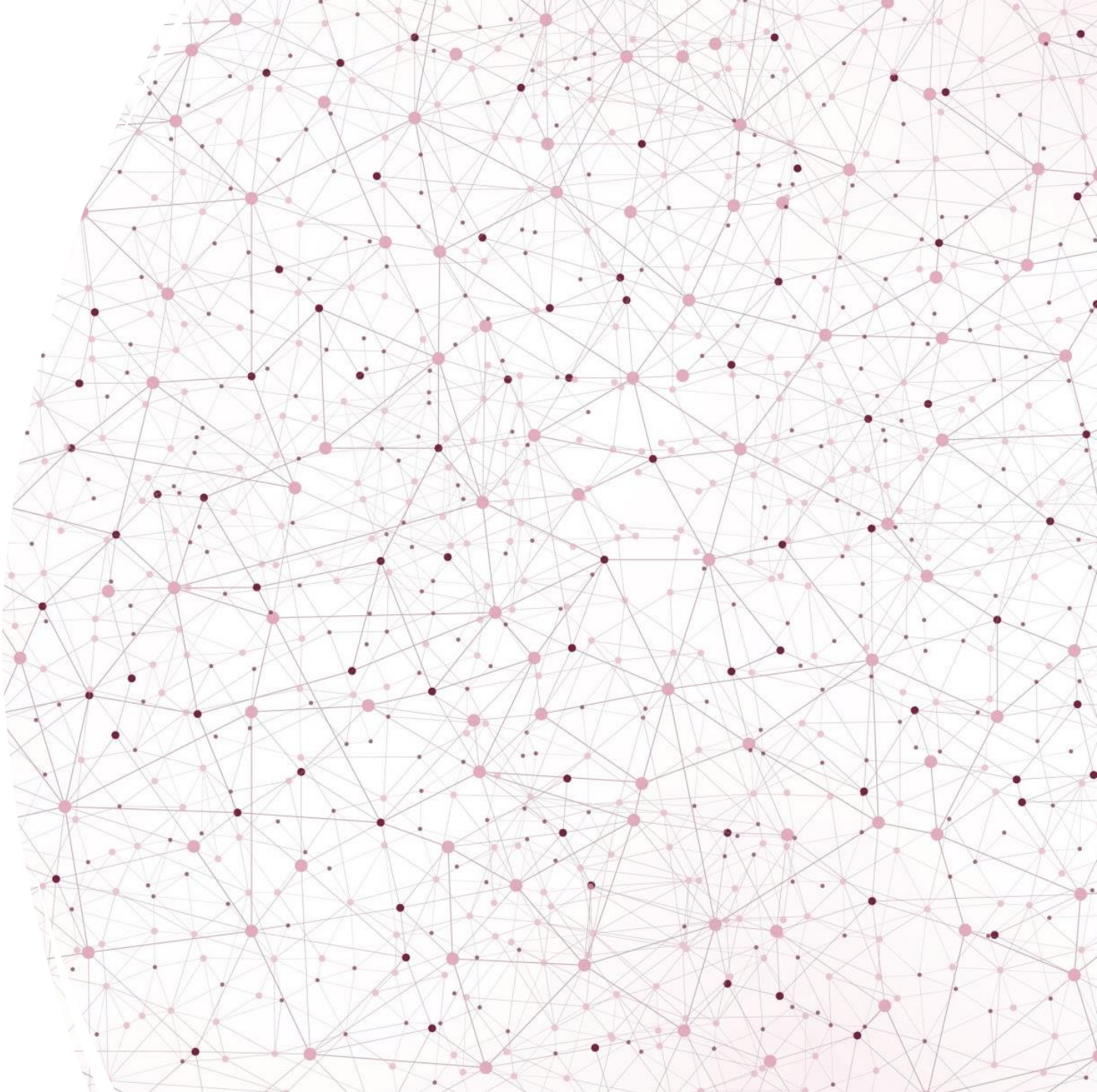
Evaluating Machine Learning Models for Accurate Cardiovascular Disease Prediction



Research Question:

How effectively can machine learning models predict the presence of cardiovascular disease based on patient health data, and which model provides the best performance?

Problem Statement:

Cardiovascular disease is one of the leading causes of death worldwide, making early detection very important. However, traditional diagnostic methods can be time-consuming, costly, and often rely heavily on medical expertise. With the increasing availability of healthcare data, machine learning offers a promising way to support faster and more accurate predictions.

Despite this potential, selecting the most effective model remains challenging due to differences in data quality, feature importance, and model performance. This project aims to develop and compare multiple machine learning models to identify the most reliable approach for predicting cardiovascular disease

Aim

To develop and compare machine learning models for accurately predicting the risk of cardiovascular disease based on patient health indicators.

Objectives:

- **Data Pre-processing**: Clean and prepare the dataset (handle outliers, missing values, and inconsistencies).
- **Feature Engineering**: Create new meaningful features (e.g., BMI, Pulse Pressure, Risk Score) to improve model performance.
- **Exploratory Data Analysis (EDA):** Analyze relationships between health factors and heart disease through statistical summaries and visualizations.
- **Model Development**: Implement and train three classification models: Logistic Regression, Random Forest, and XGBoost.
- **Hyperparameter Tuning**: Optimize each model using GridSearchCV to achieve the best performance.
- **Model Evaluation**: Compare models based on accuracy, precision, recall, and F1-score to identify the most effective predictor.
- **Feature Importance:** Identify key risk factors contributing to heart disease from the analysis. write this ion more efficient way

Dataset Overview

The dataset contains **70,000 patient records** with medical and lifestyle attributes

It is used to predict the presence of **heart disease (binary target variable)**

Key Features:

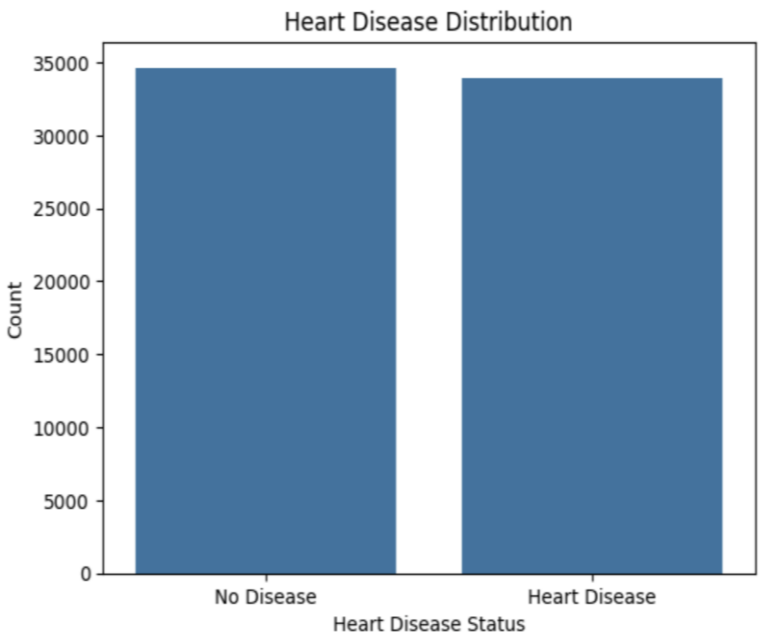
- **Demographics:** age, gender
- **Body measurements:** height, weight
- **Vital signs:** systolic BP (ap_hi), diastolic BP (ap_lo)
- **Health indicators:** cholesterol, glucose
- **Lifestyle factors:** smoking, alcohol intake, physical activity

Target variable:

- 0 → No Heart Disease
- 1 → Heart Disease

Dataset Characteristics:

- No missing values
- Balanced target classes (~50% each)
- Mix of numerical and categorical features




```
print(df.isnull().sum())
```

```
id                0
age               0
gender            0
height            0
weight            0
ap_hi             0
ap_lo             0
cholesterol       0
gluc              0
smoke             0
alco              0
active            0
Heart_Disease     0
dtype: int64
```

Data Preprocessing

Data Cleaning: The dataset was cleaned by renaming the target column to *Heart Disease* and verifying that there were no missing values. Invalid and extreme values were removed by applying conditions on blood pressure, height, and weight to ensure all values were within realistic human ranges.

Feature Engineering: New features were created to improve model performance, including BMI, pulse pressure, mean arterial pressure, and blood pressure ratio. Age was converted from days to years, and interaction features such as age–BMI and age–blood pressure were also generated.

Risk Score Creation: A combined risk score was calculated using important health and lifestyle factors such as cholesterol levels, glucose levels, smoking habits, alcohol consumption, and physical inactivity.

Feature Transformation: Variables were grouped into meaningful categories, such as BMI (underweight to obese) and age groups. Labels were also assigned to features like gender, cholesterol, glucose, smoking, and activity levels to make the data easier to interpret.

Data Preparation: The final dataset was prepared by selecting relevant features, converting categorical variables into numeric format, splitting the data into training and testing sets (70:30), and applying feature scaling using Standard-Scaler.

Exploratory Data Analysis (EDA)

Overview

- EDA was performed to understand the data, identify patterns, and analyze relationships with heart disease.

Key Observations

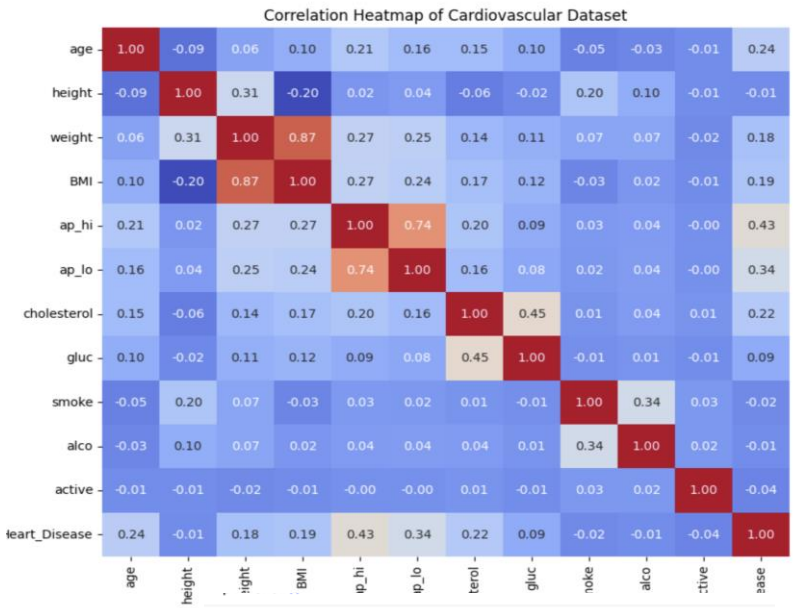
- The dataset is balanced with equal cases of heart disease and no disease. Age, BMI, and blood pressure show strong influence on heart disease.

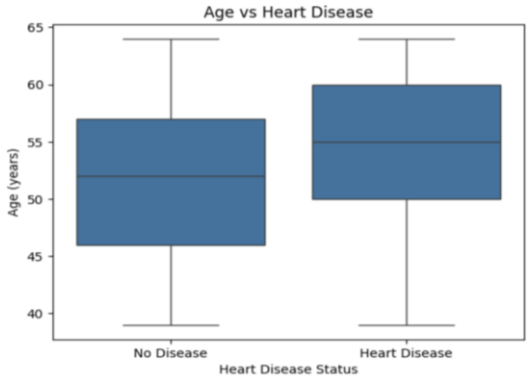
Lifestyle & Health Factors

- Higher cholesterol and glucose levels are linked to increased risk. Physical activity reduces risk, while smoking and alcohol have moderate impact.

Correlation Insights

- Strong relationships exist between blood pressure features and BMI-related variables. Age, cholesterol, and blood pressure show notable correlation with heart disease.





Models Implementation

Random Forest Classifier

Random Forest is an **ensemble machine learning algorithm** that builds multiple decision trees and combines their out to improve prediction accuracy.

Each tree is trained on a random subset of the data and features, which reduces overfitting and increases generalizatio

Handles **large datasets** and high-dimensional features efficiently.

Robust to **outliers and noise** in the data.

Can model **complex, non-linear relationships** between features.

Provides **feature importance**, helping identify key factors influencing predictions.

Hyperparameter Tuning

Used **GridSearchCV** to tune important parameters

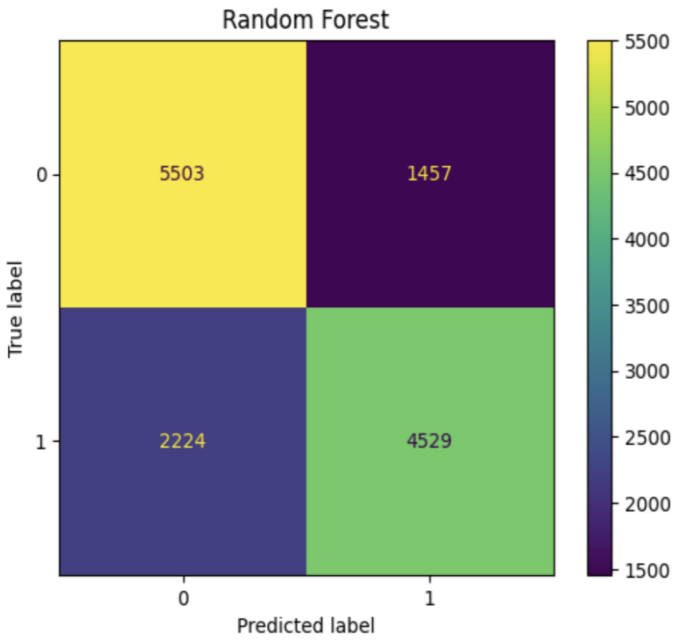
Best Parameters

n_estimators = 200

max_depth = 10

min_samples_split = 2

min_samples_leaf = 1



Logistic Regression

Definition

Logistic Regression is a **statistical classification algorithm** used to predict binary outcomes, such as presence or absence of heart disease.

It estimates the probability of an event occurring using a logistic function.

Advantages

Simple and easy to interpret.

Works well with **linearly separable data**.

Provides **probability scores** for predictions.

Fast to train and suitable for **baseline models**.

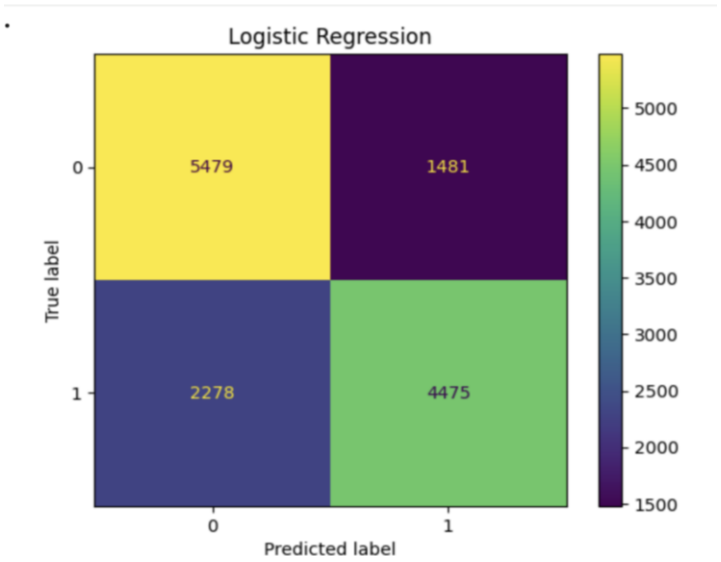
Hyperparameter Tuning & Best Parameters

Used **GridSearchCV** to tune:

C (inverse of regularization strength)
solver (algorithm for optimization)

Best Parameters Found:

C = 10
solver = liblinear



XGBoost

- **Definition**

XGBoost (Extreme Gradient Boosting) is a **powerful ensemble algorithm** that builds decision trees sequentially, where each tree corrects the errors of previous trees.

Known for high accuracy and efficiency in classification tasks.

- **Advantages**

Handles **large datasets** and **complex non-linear relationships**.

Reduces overfitting using **regularization**.

Fast and optimized for **parallel processing**.

Provides **feature importance** for interpretability.

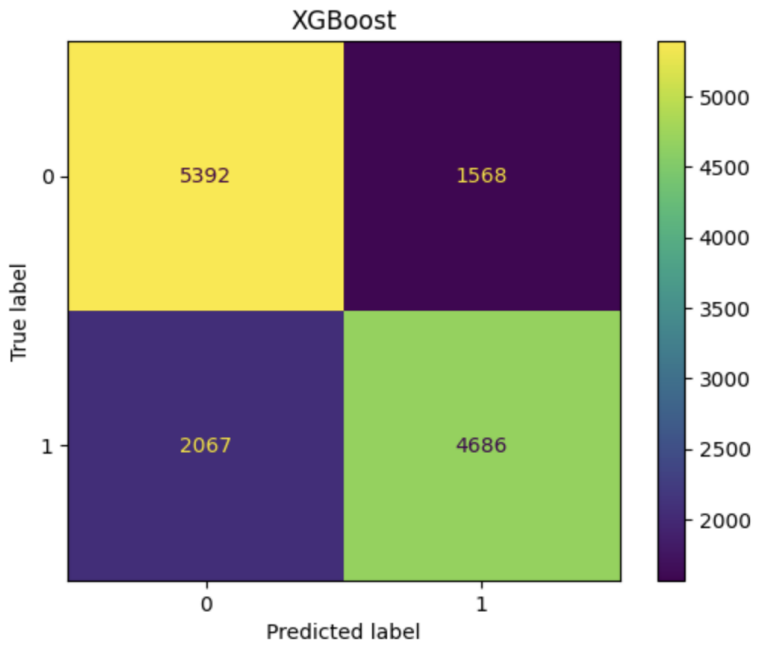
- **Hyperparameter Tuning & Best Parameters**

Used **GridSearchCV** to tune:

n_estimators (number of trees)
max_depth (maximum depth of trees)
learning_rate (step size for updates)
subsample (fraction of samples per tree)

Best Parameters Found:

n_estimators = 200
max_depth = 3
learning_rate = 0.1
subsample = 0.8



Conclusion

The project successfully developed machine learning models to predict heart disease using patient health and lifestyle data. Logistic Regression, Random Forest, and XGBoost were implemented and tuned, with Random Forest achieving the best overall performance. Key risk factors such as age, blood pressure, BMI, cholesterol, and glucose were identified, and feature engineering, including BMI, pulse pressure, risk scores, and interaction features, improved model accuracy.

These findings demonstrate that machine learning can be an effective tool for early detection of high-risk patients. By identifying critical health indicators and predicting heart disease, these models can support preventive healthcare and assist clinicians in making informed decisions.